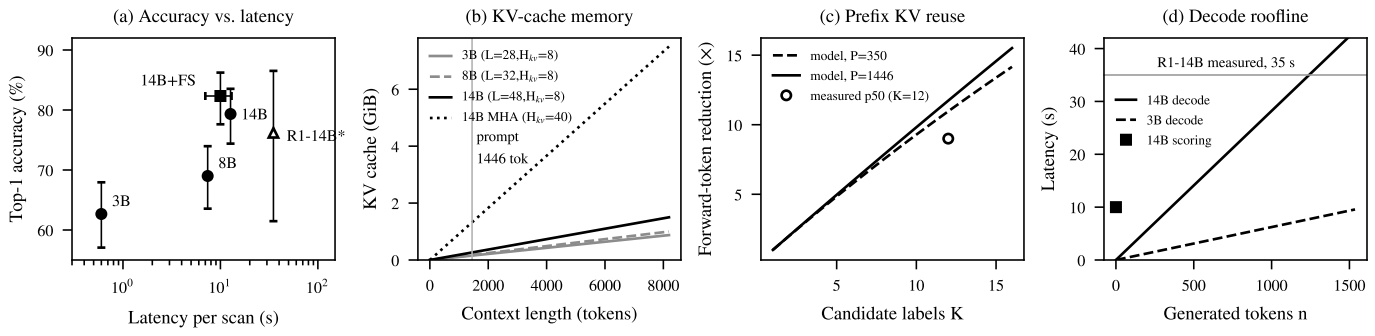
(a) Accuracy vs. latency
Top-1 accuracy (%)
90
80
70
60
14B+FS
14B
8B
3B
R1-14B*
10^0
10^1
10^2
Latency per scan (s)
(b) KV-cache memory
KV cache (GiB)
0
2
4
6
3B (L=28,H_{kv}=8)
8B (L=32,H_{kv}=8)
14B (L=48,H_{kv}=8)
14B MHA (H_{kv}=40)
prompt
1446 tok
0
2000
4000
6000
8000
Context length (tokens)
(c) Prefix KV reuse
Forward-token reduction (×)
15
10
5
model, P=350
model, P=1446
measured p50 (K=12)
5
10
15
Candidate labels K
(d) Decode roofline
Latency (s)
40
30
20
10
0
R1-14B measured, 35 s
14B decode
3B decode
14B scoring
0
500
1000
1500
Generated tokens n